

IFYPH004 2024/25

NCUK
THE UNIVERSITY CONSORTIUM

PHYSICS

INTERNATIONAL FOUNDATION YEAR (IFY)



TABLE OF CONTENTS

Syllabus Overview	2
General Learning Outcomes	3
Assessment Objectives	4
Assessment	3
Teaching, Learning and Assessment Guidance	5
Resources	12
Module Content	14
Appendix A - Exemplar Teaching Plan	36
Appendix B - Data, Formulae & Relationships	49

SYLLABUS OVERVIEW

Module Code	IFYPH004
Module Name	Physics
Programme Name	International Foundation Year
Percentage breakdown of Coursework	20%
Percentage breakdown of Examination/Test	80%
Delivery period	The syllabus will usually be delivered over two 15-week semesters
Recommended minimum teaching hours	120 (4 hours per week over 30 weeks)
Recommended minimum independent study hours	120 (4 hours per week over 30 weeks)

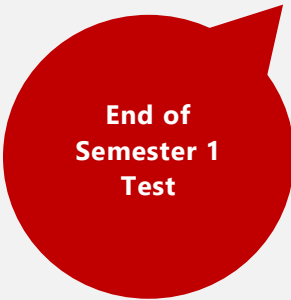
AIMS

The module aims to enable students to acquire the knowledge and understanding of physics to meet the entry requirements of NCUK Universities. Students will further develop their skills of investigation through practical experience and tutorial-based learning and will apply and practise their ICT skills in the study of physics. They will develop confidence and competence as learners, taking some personal responsibility for their own learning through directed study and reading. Students will become familiar with science-specific English language vocabulary and terminology and practise skills introduced in EAP/EAPPU/RCS lessons, together preparing them to learn effectively in an English language university.

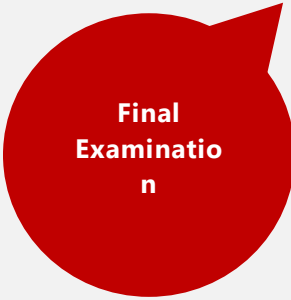
TOPICS OF STUDY

- ⇒ Mechanics
- ⇒ Materials
- ⇒ Electricity
- ⇒ Fields
- ⇒ Waves
- ⇒ Atoms

ASSESSMENT

	What is Assessed?	Topics A - V
	Duration/Word Count	1 hour 40 minutes
	Total Marks	50
	Rubric	⇒ 10 multiple choice questions (10 marks in total) ⇒ 2 short answer questions (20 marks in total) ⇒ 1 longer question from a choice of 2 (20marks in total)
	Contribution to Overall Grade	10%

	What is Assessed?	Topics A-AG
	Duration/Word Count	Report of 800-1000 words (plus calculations, diagrams, references etc.)
	Total Marks	100
	Rubric	⇒ Laboratory Report
	Contribution to Overall Grade	20%

	What is Assessed?	Topics A - AL
	Duration/Word Count	2 hours 40 minutes
	Total Marks	100
	Rubric	⇒ 10 multiple choice questions (10 marks in total) ⇒ 3 short answer questions (30 marks in total) ⇒ 3 longer questions from a choice of 5 (60 marks in total)
	Contribution to Overall Grade	70%

ASSESSMENT OBJECTIVES

	Description	End of Semester 1 Test (10%)	Coursework (20%)	Exam (70%)	Overall
AO1	Knowledge and Understanding: Demonstrate knowledge and understanding of terminology, facts, theories and concepts related to physics.	50%	~	50%	40%
AO2	Handling and Evaluating Information: Manipulate data accurately and use numerical and other information to identify patterns and trends. Give reasoned explanations for identified patterns and relationships.	50%	~	50%	40%
AO3	Practical and Investigative Skills: Demonstrate the ability to plan and carry out practical work, accurately collect and record results, and analyse and interpret data to reach a conclusion.	~	100%	~	20%

GENERAL LEARNING OUTCOMES

On successful completion of this module, a student will be able to:

Knowledge and understanding	⇒ Describe, explain and interpret phenomena and effects covered in the syllabus in terms of physical principles and concepts. ⇒ Discuss the significance of physical concepts and experimental results.
Intellectual skills	⇒ Present and interpret data as text or in tables, diagrams and graphs in appropriate and correct English. ⇒ Carry out relevant calculations, giving units where appropriate. ⇒ Use appropriate software in the analysis of data and presentation of findings.
Practical skills	⇒ Plan experimental activities, selecting appropriate techniques. ⇒ Carry out laboratory experiments safely. ⇒ Make observations and measurements with appropriate precision and record these methodically. ⇒ Interpret and discuss experimental results.
Transferable skills	⇒ Use and reference information from a variety of English language sources such as texts and the internet. ⇒ Study independently and make personal notes for revision purposes. ⇒ Write clear and concise reports in appropriate scientific English, using text, tables and graphs.

TEACHING, LEARNING AND ASSESSMENT GUIDANCE

Teachers should use a range of different learning and delivery styles in order to give students experience of the types of approach they will encounter in an NCUK University e.g. lectures, tutorials, seminars, laboratory practicals etc. Appendix A (Exemplar Teaching Plan) suggests a delivery format designed to facilitate teaching and learning in each topic. This is provided as a guide and teachers are at liberty to diverge from this plan. The Study Centre may also choose to timetable more than the minimum number of teaching hours (4 hours per week) to meet the needs and abilities of students.

Teachers should take every opportunity to use case study and data response or practice material to develop the ability of students to apply their knowledge and develop their skills of handling and evaluating information. Tutorials should involve both group and individual activities, with a strong emphasis on applying knowledge from lectures and independent study to problem-solving. It is important for all students to have opportunities to speak in English during each tutorial. Suggested activities include students being encouraged to explain in English their answers in pairs or small groups, students providing answers to the whole class whilst standing at the front of the group, student-led presentations, and group activities that require discussion and debate.

Students will have different backgrounds in the subject and it will be necessary to provide opportunities for directed independent study, so as to allow each student to develop at their own pace to reach the required level for the assessments. Activities (homework) for independent study should be set weekly. It is advisable to also consult with EAP teachers when planning homework as there may be subject tasks which can support the development of academic English (and vice-versa).

DEVELOPING INDEPENDENT LEARNING

It is important that the learners are given the guidance and support to develop from pupils within a teacher-led learning environment, to students who feel confident taking ownership of their learning through independent study. As students may have an educational background where concepts of teaching and learning are different to those prevalent in NCUK Universities, aspects of independent learning should be introduced gradually throughout the programme.

Below are suggested examples of ideas which could be used to encourage independent learning:

Acknowledging Failure and Success

A fear of failure or making mistakes can be a significant obstacle which prevents learners from being independent. It is therefore important to acknowledge the benefits of failure with regard to learning early in the module delivery. This could be explored in more depth through practical examples and calculations in which failure leads to success. Teachers should ensure that students are also praised, particularly through formative feedback, and encourage learners to demonstrate development from their past mistakes.

Classroom Reversal

Where possible, teachers might entrust the teaching, practice and revision of sub-topics or items from the syllabus to students themselves. Sharing the responsibility of teaching with students can be a more motivating way to encourage them to take their first steps into independent study than being asked to

study independently for purely personal reasons. It also provides a strong reason why independent learning should be carried out (in this case, to support their peers) and a visible outcome of their independent work (the performance of their peers).

A demonstration of possible student-led tasks and activities can be a useful way to introduce this concept within the classroom. A clear schedule for student-led sessions can also help to make it routine. Sessions could be led by groups initially before asking students to lead them individually when they have grown in confidence.

PAST ASSESSMENTS

Many learners can find it reassuring to revisit past assessments to support their preparation for formative and summative assessments. A selection of past Coursework, End of Semester 1 Test and Final Examination papers are available for this purpose via the NCUK Teachers' Hub. It is important that students are guided to sections and questions which will support the meeting of current learning outcomes or that past assessments are appropriately adapted to ensure their fit to the current syllabus.

While classroom time can be allocated to the review of and use of past papers, students should also have the opportunity to make use of these for independent study. Teachers should work with colleagues to decide which papers may be kept aside for use in formative assessment or class time and which to release to students for independent study.

There are a number of ways past assessments can be used to support learning in addition to recalling knowledge. These include:

- ⇒ An exploration of incorrect or incomplete answers which could highlight gaps in understanding and help to inform future teaching and learning
- ⇒ Identifying whether the correct structure is being used to answer extended answers
- ⇒ Practicing answers to certain questions within particular time constraints
- ⇒ Distinguishing different command words, such as 'explain', 'describe' and 'evaluate'
- ⇒ Allowing students to notice weaknesses in their own knowledge or answers by allowing them to self-mark with the assessment criteria
- ⇒ Allowing students to establish a greater familiarity with the assessment criteria and what they are expected to do to achieve a certain grade

SUPPORTING ENGLISH LANGUAGE DEVELOPMENT

It is important for the development of academic literacies that the delivery of the subject material is integrated with the English for Academic Purposes (EAP) module (or EAP for Proficient Users or Research and Communication Skills modules where appropriate). Regular communication between the subject module teacher(s) and the EAP, EAPPU or RCS teacher(s) will provide a basis on which to support and guide students. Students will benefit from collaborative activities, where the subject module and EAP/EAPPU/RCS teachers jointly deliver classes in relation to activities such as essay or report writing style and using academic sources.

There are also a number of simple tasks and approaches which can be built into this module which can support language development. These include:

Introducing New Vocabulary

When students are introduced to new vocabulary in class, ensure that these words are not introduced in isolation, but in a chunk or phrase within a wider understood context. As an example, if students were given the following extract from the Economist (2021), they may be unfamiliar with the word 'ailments':

The cholera pandemic of the early 1830s hit France hard. It wiped out nearly 3% of Parisians in a month, and hospitals were overwhelmed by patients whose ailments doctors could not explain.

If this word was then to be explored and recorded, it would be better to look at this chunk:

...hospitals were overwhelmed by patients whose **ailments** doctors could not explain.

By giving students a clear context, it is easier to initially discern meaning while also making it more memorable for revision. By keeping the word within a larger chunk, there are also a number of grammatical and lexical relationships which can be established by learners:

- ⇒ Common collocations; explain ailments, a patient's ailments,
- ⇒ The inclusion of 'ailments' in the semantic field of medicine / health.
- ⇒ The determiner 'whose' implies that 'ailments' is a noun and the 's' that it is a plural, countable noun.
- ⇒ Wider examples of past passive and defining relative clauses.

Although it should not be expected that this level of exploration is conducted within a subject module, highlighting new vocabulary in chunks does give language learners a greater opportunity to notice features of language at their own speed. It also provides a grammatical template which students can build on when writing their own answers.

Highlighting Stressed Syllables

One way to support the development of students spoken fluency and overall rhythm is to draw attention the stressed syllable within key, subject-specific vocabulary. This can be done very simply on the board in a number of ways and you may wish to discuss this with the EAP/RCS teacher(s) at your Study Centre in order to align your preferred method(s). As you may be students' main model for how to pronounce subject-specific vocabulary, it is essential they have a clear record which they can refer to during independent study.

Encouraging Functional & Discursive Language

When completing written and spoken tasks, such as group discussions or the presentation of ideas within class, you may wish to encourage students to use functional or discursive language that will help them to structure and explain their ideas with greater coherence and clarity. For instance, this may include language for agreeing /disagreeing, clarifying what they have said when they are unclear, or how to highlight a particular point. This could be achieved by:

- ⇒ Asking students to take out, look at and use language explored for this purpose within their EAP classes. Collaboration with the EAP teacher(s) to decide when in the module particular language will be covered and for what purpose will greatly support students' language development.

- ⇒ Make functional and discursive language relevant to your subject available in the classroom for students to use. Adding useful language to walls or 'vocabulary mats' on desks can remind students of language available to them while also encouraging independence.

Allowing Learners Time to Speak

Although some students may be reluctant to speak in class, it is also important to remember that less proficient speakers require time to construct utterances and may need some time to prepare themselves. While silence in a classroom can sometimes feel awkward, try to avoid filling this silence with teacher talk time, particularly when you are expecting answers from students. You may want to prepare students to speak by giving them advance warning, for example, by giving the class time to make notes before being asked to speak. Alternatively, you may wish to select students to answer particular questions in advance.

Referencing

As part of study for the EAP, EAPPU or RCS module, students will learn the Harvard referencing system. Subject teachers should ensure that students carry this learning into the coursework assessments for this module; see the 'Resources' section of this document for details of the referencing guide recommended by NCUK. For further information about referencing and citation, please consult the EAP or RCS syllabus (as relevant) for the texts and online resources recommended by NCUK.



FORMATIVE ASSESSMENT

It is important that students are given the opportunity to engage in and submit formative assessments and receive feedback on this work. Formative assessment should be designed to inform students of their progress and enable them to develop and practice coursework and examination skills.

In order to assist teachers in setting formative assessment, NCUK provides the following for this module:

- ⇒ Formative coursework assessment – details of laboratory practicals that can be completed for formative assessment are given in the Physics Laboratory Practical Guide which is available to download from the NCUK Teachers' Hub. Students should have the opportunity to undertake at least one laboratory practical under assessed conditions in Semester 1 to prepare them for the summative laboratory report in Semester 2. This assessment will provide an opportunity for students to receive feedback that can be applied to the summative coursework in Semester 2 and also help teachers to generate predicted grades. When marking this assessment, teachers should use the NCUK issued Assessment Criteria given in Appendix A of the Physics Laboratory Practical Guide. Feedback should be given to students using the feedback sheet in Appendix B of the Physics Laboratory Practical Guide.
- ⇒ Exemplar assessments – a full set of exemplar assessments with mark schemes (End of Semester 1 Test and Final Examination) are downloadable from the Teachers' Hub. These can be used as mock assessments for students throughout the academic year.
- ⇒ Past exam questions – a set of past examination papers is accessible for teachers via the Teachers' Hub. Teachers can generate their own papers for use as practice exams, pick out specific questions to work through with students in class, or set as homework.

In addition to the above, teachers should set regular homework tasks that allow students to understand their strengths and weaknesses and should provide advice to students on how to improve in areas of weakness.

Beyond developing students' skills, the crucial element of formative assessment is the feedback provided by the teacher.



SUMMATIVE ASSESSMENT

The breakdown of summative assessment for this module and the rubrics for each component are given on page 3 of this document. All summative assessments are provided by NCUK.

End of Semester 1 Test

The End of Semester 1 Test will cover topics from Semester 1 only and will take place under examination conditions. Teachers at the Study Centre will be responsible for marking the End of Semester 1 Tests in accordance with the NCUK issued mark scheme.

Coursework

The coursework assessment should be completed during Semester 2. Teachers should consult the Physics Laboratory Practical Guide to identify a suitable practical for assessment. Teachers are free to set their own submission deadline. Students will submit their coursework online via NCUK's assessment platform. All assessments will be subject to a plagiarism check using Turnitin.

Teachers at the Study Centre will be responsible for marking coursework in accordance with the NCUK issued mark scheme and the assessment criteria given in Appendix A of the Physics Laboratory Practical Guide. Students should be given sight of the assessment criteria when they are undertaking the practical task as it helps to inform them of what is expected at each grade level. When marking, teachers will either complete the marking and feedback form given in Appendix B and upload this to the NCUK learning and assessment platform or complete the marking rubric presented within the platform.

Examination

The examination will cover topics from both semesters. Teachers at the Study Centre will be responsible for marking the examination in accordance with the NCUK issued mark scheme.

- ⇒ An NCUK approved calculator is permitted in the examination (refer to NCUK policy 'Calculator Regulations').
- ⇒ It is the Study Centre's responsibility to provide graph paper for students for the End of Semester 1 Test and Final Examination.
- ⇒ A formula booklet ('Data, Formulae and Relationships' - refer to Appendix B of this syllabus) will be included in the question papers for the End of Semester 1 Test and Final Examination. Study Centres are not required to provide this separately for students.

Supporting Documents

It is essential that coursework and examinations are administered in accordance with NCUK regulations. Please refer to the following sections of the NCUK Academic Handbook for details:

- ⇒ Coursework Administration and Regulations
- ⇒ Centre Marking and Recording Results

- ⇒ Academic Misconduct Policy
- ⇒ Examination Administration

RESOURCES



Core Textbook

Breithaupt, J. (2015), *AQA Physics*, 2nd Ed. Oxford University Press

ISBN: 9780198351870 (Print)

ISBN: 9780198378495 (eBook)

Johnson, K., Hewett, S., Holt, S & Miller, J. (2015) *Advanced Physics for You*. 2nd Ed. Oxford University Press

ISBN: 9781408527375 (Print)

ISBN: 9780198355991 (eBook)

Adams, S. & Allday, J. (2013), *Advanced Physics*, 2nd Ed. Oxford University Press

ISBN: 9780198392927 (Print)



Further Reading

Lowe, T.L. & Rounce, J.F. (2002), *Calculations for A-level Physics*, 4th Ed. Nelson Thomas

ISBN: 9780748767489 (Print)

Breithaupt, J. (2015). *AQA Physics: A level year 1 and AS*, 2nd Ed. Oxford University Press

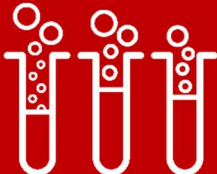
ISBN: 9780198351863 (Print)

ISBN: 9780198378488 (eBook)

Breithaupt, J. (2015), *AQA Physics: A level year 2*, 2nd Ed. Oxford University Press

ISBN: 9780198357728 (Print)

ISBN: 9780198378501 (eBook)

 Useful Websites	A good selection of interactive simulations is available at: http://phet.colorado.edu An extensive website covering all aspects of physics with extensive cross referencing: http://hyperphysics.phy-astr.gsu.edu/hbase/hph.html A selection of revision websites: http://www.a-levelphysicstutor.com http://alevelphysics.org.uk/ Virtual experiments on YouTube: A-level Physics Required Practicals - YouTube Practical Endorsement Tasks - A Level Physics - YouTube
 Referencing	Refer to the Harvard Referencing Guide in the NCUK Academic Handbook for NCUK guidelines on this, though using online Harvard Reference Guides to support assessment writing can also be beneficial. When referencing, the main objectives are clarity, consistency, accuracy of key information and ability to locate the source.
 Teacher Resources	NCUK Teachers' Hub – accessible via the NCUK learning and assessment platform – please contact your Academic Manager for access.
 Lab Requirements	Please see the accompanying 'IFY Physics Laboratory Practical Guide' for details of general lab requirements and specific equipment requirements for formative and summative laboratory practicals.

MODULE CONTENT

All topics must be covered. Appendix A provides an Exemplar Teaching Plan which indicates the proportion of time to be spent on each topic and activities to support student learning.

The tables below detail the content of the module. The module is split into 6 major topics (Mechanics, Materials, Electricity, Fields, Waves and Atoms). Each topic is made up of a number of sub-topics. A set of specific learning outcomes (SLOs) are given for each sub-topic.

TOPIC A – MECHANICS: PHYSICAL QUANTITIES AND UNITS		
CONTENT	SLOs	
⇒ Physical quantities and units ⇒ Use of the SI system	A1	Explain that physical quantities have a magnitude and unit which together can be represented by an approved italic symbol.
	A2	Use the SI base and derived units appropriate to this syllabus, along with appropriate suffixes for multiples and submultiples.

TOPIC B – MECHANICS: SCALARS AND VECTORS		
CONTENT	SLOs	
⇒ Scalars and vectors ⇒ Resolution and addition of vectors ⇒ Vector triangles	B1	Explain that scalar quantities have magnitude and vector quantities have magnitude and direction. Give examples of each quantity included in this syllabus.
	B2	Draw and use a vector triangle to determine the resultant of two co-planar vectors.
	B3	Resolve a vector into two perpendicular components and determine the resultant of two perpendicular vectors, either by drawing or calculation.

TOPIC C – MECHANICS: KINEMATICS		
CONTENT	SLOs	
⇒ Kinematics ⇒ Graphical application and use ⇒ Equations of motion ⇒ Simple experiments	C1	Represent displacement, velocity and acceleration in graphical form.
	C2	Determine displacement, velocity and acceleration from the gradients and areas of graphs.
	C3	Select and use the equations of motion for constant acceleration for a body moving in one or two dimensions: ⇒ $v = u + at$ ⇒ $s = \frac{1}{2}(u + v)t$ ⇒ $s = ut + \frac{1}{2}at^2$ ⇒ $v^2 = u^2 + 2as$
	C4	Determine equations of motion for constant acceleration for a body moving in one or two dimensions from a velocity - time graph.
	C5	Describe an experiment to determine the acceleration due to gravity of a falling body.

TOPIC D – MECHANICS: DYNAMICS		
CONTENT	SLOs	
⇒ Dynamics ⇒ The newton ⇒ Friction and drag	D1	Define the <i>newton</i> .
	D2	Use the equations of motion and $F = ma$ to solve problems involving the motion of objects.
	D3	Explain that the weight of an object is the gravitational force acting on the object and use the equation $W = mg$
	D4	Explain that objects in motion may be subject to the resistive forces <i>friction</i> and <i>drag</i> , including the motion of objects in one dimension and two dimensions in the presence of friction and drag, and explain the term <i>terminal velocity</i> .

TOPIC E – MECHANICS: NEWTON’S LAWS OF MOTION		
CONTENT	SLOs	
⇒ Newton's laws of motion ⇒ Linear momentum	E1	State and use each of Newton's three laws of motion.
	E2	Define linear momentum as the product of mass and velocity and appreciate the vector nature of momentum.
	E3	Define net force on a body as equal to rate of change of its momentum.
	E4	Select and apply the equation $F = \Delta p / \Delta t$

TOPIC F – MECHANICS: STATICS		
CONTENT	SLOs	
⇒ Statics ⇒ Centre of gravity ⇒ Moments ⇒ Couples and torques ⇒ Triangle of forces	F1	Explain that the centre of gravity of an object is the single point through which the full weight of a body appears to act. Describe a simple experiment to determine the centre of gravity of a body.
	F2	Define <i>moment of force</i> . Apply the principle of moments to solve problems involving single support, two support and cantilever systems in equilibrium.
	F3	Define and apply the terms <i>couple</i> and <i>torque</i> .
	F4	Draw and use a triangle of forces to solve problems of the equilibrium of coplanar forces.

TOPIC G – MECHANICS: WORK AND CONSERVATION OF ENERGY		
CONTENT	SLOs	
⇒ Work and conservation of energy ⇒ The joule ⇒ Conservation of energy ⇒ Pressure	G1	Define the <i>work</i> done by a force.
	G2	Define the <i>joule</i> .
	G3	Calculate the work done by a force using: ⇒ $W = Fx$ and ⇒ $W = Fx \cos \theta$
	G4	State and apply the principle of conservation of energy.
	G5	Solve problems involving work done and the transfer of energy.
	G6	State and use the equation for pressure: ⇒ $p = F/A$

TOPIC H – MECHANICS: KINETIC AND POTENTIAL ENERGIES		
CONTENT	SLOs	
⇒ Kinetic and potential energies. ⇒ Power ⇒ Efficiency	H1	State and use the equation for kinetic energy: $\Rightarrow E_k = \frac{1}{2}mv^2$
	H2	Define <i>potential energy</i> and understand that the gravitational potential energy of a mass near the Earth's surface is given by mgh .
	H3	Use the law of conservation of mechanical energy in calculations for particles moving under gravity.
	H4	Define power as the rate of work done.
	H5	Define the <i>watt</i> .
	H6	Calculate power when solving problems.
	H7	Use the relationship for efficiency: $\eta = \frac{\text{useful output energy}}{\text{total input energy}} \times 100\%$
	H8	Explain that η is always $< 100\%$ because of heat losses.

TOPIC I – MECHANICS: COLLISIONS		
CONTENT	SLOs	
⇒ Collisions ⇒ Elastic and inelastic collisions	I1	State the principle of conservation of momentum and apply the principle to solve problems when bodies interact in one dimension.
	I2	Define <i>a perfectly elastic collision</i> and <i>an inelastic collision</i> .
	I3	Explain that whilst the momentum of a system is always conserved in the interaction between bodies, some change in kinetic energy usually occurs.

TOPIC J – MECHANICS: CIRCULAR MOTION		
CONTENT	SLOs	
⇒ Circular motion ⇒ Centripetal force and acceleration ⇒ Linear velocity in circular motion	J1	Define the <i>radian</i> and convert angles from degrees into radians and vice versa.
	J2	Explain that for an object moving at constant speed in a circular path, the net force acting on the object is perpendicular to the velocity of the object and is directed towards the centre of the circle.
	J3	Explain what is meant by centripetal acceleration and centripetal force.
	J4	Select and use the equations for circular motion: ⇒ $v = 2\pi r/T = \omega r$ ⇒ $a = v^2/r$

TOPIC K – MECHANICS: SIMPLE HARMONIC MOTION (SHM) AND RESONANCE		
CONTENT	SLOs	
⇒ Simple harmonic motion (SHM) and resonance. ⇒ Periodic time and frequency ⇒ Angular velocity ⇒ Acceleration ⇒ Damping	K1	Define and use the terms <i>displacement</i> , <i>amplitude</i> , <i>period</i> , <i>frequency</i> , <i>angular frequency</i> and <i>phase difference</i> .
	K2	Select and use the equation for period: ⇒ $T = 1/f = 2\pi/\omega$
	K3	Define <i>simple harmonic motion</i> .
	K4	Select and use the SHM equations: ⇒ $a = -\omega^2 x$ ⇒ $x = A \sin \omega t$ ⇒ $v = A \omega \cos \omega t = \pm \omega \sqrt{A^2 - x^2}$
	K5	Explain that the period of an object with SHM is independent of its amplitude.
	K6	Describe, with graphical illustrations, the changes in displacement, velocity and acceleration during SHM.
	K7	Describe and explain the interchange between kinetic and potential energy during SHM.
	K8	Describe the effects of damping on oscillations.
	K9	Describe graphically how the amplitude of a forced oscillation changes with frequency near to the natural frequency of the system and the concept of resonance.

TOPIC L – MATERIALS: SOLID, LIQUID AND GAS		
CONTENT	SLOs	
⇒ Solid, liquid and gas ⇒ Pressure and density ⇒ Changes of state ⇒ Simple kinetic models ⇒ Brownian motion	L1	Define and use the concept of density $\rho = m/V$
	L2	Derive and use, from the definitions of pressure and density, the equation for hydrostatic pressure $\Delta p = \rho g \Delta h$
	L3	Describe solids, liquids and gases in terms of the spacing, ordering and motion of atoms or molecules.
	L4	Use graphs of temperature against energy to illustrate changes of state.
	L5	Describe a simple kinetic model for gases.
	L6	Describe an experiment that demonstrates Brownian motion and discuss the evidence for the movement of molecules provided by such an experiment.
	L7	Define the term <i>pressure</i> and use the kinetic model to explain the pressure exerted by gases.

TOPIC M – MATERIALS: ELASTICITY		
CONTENT	SLOs	
⇒ Elasticity ⇒ Hooke's law ⇒ Stress, strain and Young's modulus ⇒ Elastic and plastic deformation	M1	Describe tensile and compressive deformation of solids.
	M2	Describe the behaviour of springs and wires in terms of force, extension, elastic limit, Hooke's law.
	M3	State and apply to springs and wires the equation $F = -kx$
	M4	Define and use the terms <i>stress</i> , <i>strain</i> , <i>Young modulus</i> , <i>elastic limit</i> and <i>tensile strength</i> .
	M5	Describe an experiment to determine the Young modulus of a metal wire.
	M6	Define the terms <i>elastic deformation</i> and <i>plastic deformation</i> of a material.
	M7	Describe the features of the force against extension graphs for typical ductile materials.
	M8	Select and use the equations $E_p = \frac{1}{2}Fx = \frac{1}{2}kx^2$ for a material deformed within its limit of proportionality.

TOPIC N – MATERIALS: THERMAL PROPERTIES OF MATERIALS		
CONTENT	SLOs	
⇒ Thermal properties of materials ⇒ Specific heat capacity ⇒ Specific latent heat	N1	Define <i>specific heat capacity</i> .
	N2	Select and use the equation $E = mc\Delta\theta$
	N3	Describe an experiment to determine the specific heat capacity of a solid or a liquid.
	N4	Describe latent heat of fusion as energy required for increasingly energetic molecules to weaken molecular bonds.
	N5	Describe latent heat of vaporisation as energy required for increasingly energetic molecules to overcome molecular bonds.
	N6	Select and use the equation $E = L\Delta m$

TOPIC O – MATERIALS: THERMAL PHYSICS		
CONTENT	SLOs	
⇒ Thermal physics ⇒ Internal energy ⇒ Kelvin temperature scale	O1	Define internal energy of a system as the sum of the kinetic and potential energies of the molecules in the system and explain that a rise in temperature of a body leads to an increase in its internal energy.
	O2	Explain that a change of state for a substance leads to a change in its internal energy but to no change in temperature.
	O3	Explain that thermal energy is transferred from a region of higher temperature to a region of lower temperature and that regions of equal temperature are in thermal equilibrium.
	O4	Explain that the thermodynamic (Kelvin) temperature scale is an absolute scale of temperature.
	O5	State that absolute zero is the temperature at which a substance has minimum internal energy.
	O6	Convert temperatures between Kelvin and degrees Celsius: ⇒ $T(K) = \theta(^{\circ}C) + 273.2$

TOPIC P – MATERIALS: IDEAL GASES		
CONTENT	SLOs	
⇒ Ideal gases ⇒ Boyle's law ⇒ Kinetic theory of gases ⇒ Ideal gas equations	P1	State Boyle's law, $pV = \text{constant}$
	P2	State and use the equation: ⇒ $pV/T = \text{constant}$
	P3	State the basic assumptions of the kinetic theory of gases.
	P4	Interpret (qualitatively) the pressure of a gas in terms of molecular collisions.
	P5	Interpret (qualitatively) the temperature of a gas in terms of molecular energies.
	P6	State that the number of particles in one mole of any substance is equal to the Avogadro constant.
	P7	Understand that one mole of a substance has a mass equal to the atomic/molecular mass expressed in grams.
	P8	Select and use the equations: ⇒ $pV = nRT$ ⇒ $pV = NkT$

TOPIC Q – ELECTRICITY: CHARGE AND CURRENT		
CONTENT	SLOs	
⇒ Charge and current ⇒ Conventional current ⇒ The coulomb ⇒ Use of an ammeter	Q1	Explain that an electric current is a net flow of charged particles.
	Q2	Explain that electric current in a metal is due to the movement of electrons, and in an electrolyte is due to the movement of ions.
	Q3	Explain what is meant by conventional current and electron flow.
	Q4	Define the <i>coulomb</i> and use the equation: ⇒ $\Delta Q = I\Delta t$
	Q5	Describe how an ammeter may be used to measure the current in a circuit.

TOPIC R – ELECTRICITY: RESISTANCE		
CONTENT	SLOs	
⇒ Resistance ⇒ The ohm ⇒ <i>I</i>-<i>V</i> characteristics of simple components ⇒ Resistivity	R1	Define <i>resistance</i> and the <i>ohm</i> .
	R2	State Ohm's law and use the equation for resistance $R = V/I$
	R3	Describe the <i>I</i> - <i>V</i> characteristics of a resistor at constant temperature, a filament lamp and a light-emitting diode (LED).
	R4	Describe an experiment to obtain the <i>I</i> - <i>V</i> characteristics of a resistor at constant temperature, a filament lamp and an LED.
	R5	Define resistivity of a material and use the equation $R = \rho L/A$

TOPIC S – ELECTRICITY: CAPACITANCE		
CONTENT	SLOs	
⇒ Capacitance ⇒ Energy in a charged capacitor ⇒ Discharge of a capacitor through a resistor ⇒ Time constant	S1	Define <i>capacitance</i> and the <i>farad</i> .
	S2	State and use the equation $C = Q/V$
	S3	Explain that the area under a potential difference against charge graph is equal to energy stored by a capacitor.
	S4	Select and use the equations $W = \frac{1}{2}QV$ and $W = \frac{1}{2}CV^2$ for a charged capacitor.
	S5	Draw graphs that show the variation with time of potential difference, charge and current for a capacitor discharging through a resistor.
	S6	Define the time constant of a circuit and use the equation $\tau = RC$
	S8	State and use the exponential discharge equations: ⇒ $Q = Q_0 e^{-t/\tau}$ ⇒ $I = I_0 e^{-t/\tau}$ ⇒ $V = V_0 e^{-t/\tau}$
	S9	Use graphs of $\ln(Q)$, $\ln(I)$ or $\ln(V)$ against t to determine τ .

TOPIC T – ELECTRICITY: E.M.F AND P.D.		
CONTENT	SLOs	
⇒ E.M.F and P.D. ⇒ Use of a voltmeter ⇒ Internal resistance	T1	Define <i>potential difference</i> (p.d.) and the <i>volt</i> .
	T2	Describe how a voltmeter may be used to determine the p.d. across a component.
	T3	Define the <i>electromotive force</i> (e.m.f.) of a power supply.
	T4	Describe the difference between e.m.f. and p.d. in terms of energy transfer.
	T5	Explain that all sources of e.m.f. have an internal resistance.
	T6	Select and use the equations: ⇒ $e.m.f. = I(R + r)$ ⇒ $e.m.f. = V + Ir$

TOPIC U – ELECTRICITY: POWER		
CONTENT	SLOs	
⇒ Electrical power ⇒ Electrical work	U1	Select and use the equations for power: ⇒ $P = IV$ ⇒ $P = I^2R$ ⇒ $P = V^2/R$
	U2	Select and use the equations for work: ⇒ $W = IVt$ ⇒ $W = QV$

TOPIC V – ELECTRICITY: D.C. CIRCUITS		
CONTENT	SLOs	
⇒ D.C. circuits ⇒ Approved circuit symbols ⇒ Kirchhoff's laws ⇒ Resistors in series and parallel ⇒ Potential divider ⇒ Capacitors in series and parallel	V1	Interpret and draw circuit diagrams using circuit symbols in current SI standards.
	V2	State Kirchhoff's first law and describe it as a consequence of conservation of charge.
	V3	State Kirchhoff's second law and describe it as a consequence of conservation of energy.
	V4	Apply Kirchhoff's first and second laws to circuits.
	V5	Select and use the equations for the total resistance of two or more resistors in series and in parallel and solve problems with resistors in these circuits.
	V6	Use the potential divider equation: ⇒ $V_{\text{out}} = [R_2 / (R_1 + R_2)] \times V_{\text{in}}$ Explain how a potential divider circuit can be used to produce a variable p.d.
	V7	Select and use the equations for the total capacitance of two or more capacitors in series and in parallel and solve problems with capacitors in series and parallel circuits.
	V8	Solve circuit problems with one or more sources of e.m.f.

TOPIC W – FIELDS: ELECTRIC FIELDS		
CONTENT	SLOs	
⇒ Electric fields ⇒ Electric field strength ⇒ Coulomb's law ⇒ Electric potential ⇒ Electric field lines and equipotentials	W1	State that electric fields are created by electric charges.
	W2	Define electric field strength as force per unit positive charge and use the equation: ⇒ $E = F/Q$.
	W3	Select and use the equation for Coulomb's law: ⇒ $F = Qq/(4\pi\epsilon_0 r^2)$
	W4	Select and use the equation for the electric field strength of a point charge: ⇒ $E = Q/(4\pi\epsilon_0 r^2)$
	W5	Define <i>electric potential at a point</i> and use the equation: ⇒ $V = Q/(4\pi\epsilon_0 r)$
	W6	State that a uniform electric field exists between oppositely charged parallel plates.
	W7	Select and use the equation for the field strength between charged parallel plates: ⇒ $E = V/d$
	W8	Draw the electric field lines and equipotentials for a point charge, a spherically symmetric charge distribution, the region between two parallel plates and an electric dipole.
	W9	Explain that a charged particle with charge q is accelerated by a potential difference ΔV and show that $q\Delta V = \frac{1}{2}mv^2$
	W10	Use the equation $eV = \frac{1}{2}mv^2$ for electrons and other charged particles.

TOPIC X – FIELDS: MAGNETIC FIELDS		
CONTENT	SLOs	
⇒ Magnetic fields ⇒ Magnetic flux density ⇒ Fleming's left-hand rule ⇒ Forces on charged particles and currents in a magnetic field ⇒ Magnetic field lines	X1	Define <i>magnetic field B</i> or <i>magnetic flux density B</i> (and its unit the <i>tesla</i>) in terms of the force on a current carrying conductor.
	X2	State and apply Fleming's left-hand rule to determine the direction of force on a current carrying conductor placed at right angles to a magnetic field.
	X3	Select and use the equations: ⇒ $F = BIL$ ⇒ $F = BIL \sin \theta$
	X4	Select and use the equation for the force on a charged particle moving in a direction perpendicular to a magnetic field: ⇒ $F = Bqv$
	X5	Derive and use the equation of motion for a charged particle moving perpendicular to a magnetic field: ⇒ $Bqv = mv^2/r$
	X6	Draw the magnetic field lines for a long straight current-carrying wire, a long solenoid, a bar magnet and the Earth.

TOPIC Y – FIELDS: ELECTROMAGNETISM		
CONTENT	SLOs	
⇒ Electromagnetism ⇒ Torque on a coil ⇒ D.C. motor ⇒ Magnetic flux and the weber ⇒ Faraday's law ⇒ Lenz's law ⇒ Fleming's right-hand rule ⇒ Simple a.c. generator	Y1	State and use the expression $T = BIAN$ for the torque on a coil placed so that its plane is parallel to a uniform magnetic field B .
	Y2	Describe the operation of a d.c. motor with an iron core, radial magnetic field and commutator.
	Y3	Define <i>magnetic flux</i> and the <i>weber</i> .
	Y4	State and use the equation for magnetic flux linking N turns of a coil placed so that the perpendicular to its plane is at an angle θ to a magnetic field B : ⇒ $N\Phi = NBA\cos\theta$
	Y5	State and apply Faraday's law of electromagnetic induction: ⇒ e.m.f. = $-N(\Delta\Phi / \Delta t)$
	Y6	State and apply Lenz's law to determine the direction of induced e.m.f.
	Y7	State and apply Fleming's right-hand rule to determine the direction of the current induced in a conductor moving at right angles to a magnetic field.
	Y8	Describe the operation of a simple a.c. generator.

TOPIC Z – FIELDS: GRAVITY		
CONTENT	SLOs	
⇒ Gravity ⇒ Gravitational field strength ⇒ Gravitational field lines ⇒ Gravitational potential ⇒ Newton's law of gravitation ⇒ Circular orbits ⇒ Kepler's third law ⇒ Geostationary orbits	Z1	Describe how a mass creates a gravitational field in the space around it.
	Z2	Define gravitational field strength as force per unit mass.
	Z3	Use gravitational field lines to represent a gravitational field.
	Z4	Define <i>gravitational potential at a point</i> and use $V = W/m$ for the gravitational potential in the field due to a small object of mass m .
	Z5	State and use Newton's law of gravitation: ⇒ $F = GMm/r^2$
	Z6	Select and apply the equation $g = GM/r^2$ for the gravitational field strength of a point mass.
	Z7	Determine the mass of the Earth or another body from its gravitational field strength.
	Z8	Explain that close to the Earth's surface the gravitational field strength is uniform and equal to the acceleration of free fall.
	Z9	Analyse circular orbits by relating the gravitational force to the centripetal force and so derive the period of an orbit for planets and satellites as $T^2 = 4\pi^2 r^3 / (GM)$.
	Z10	Apply Kepler's third law to solve problems: ⇒ $T^2 \propto r^3$
	Z11	Define the <i>geostationary orbit</i> of a satellite and use this concept in calculations.

TOPIC AA – WAVES: WAVE MOTION		
CONTENT	SLOs	
⇒ Wave motion ⇒ Longitudinal and transverse waves ⇒ Wave equation	AA1	Describe and distinguish between progressive longitudinal and transverse waves.
	AA2	Define and use the terms <i>displacement, amplitude, wavelength, period, phase difference, frequency</i> and <i>speed</i> of a wave.
	AA3	Use the wave equation $v = f\lambda$

TOPIC AB – WAVES: ELECTROMAGNETIC WAVES		
CONTENT	SLOs	
⇒ Electromagnetic waves ⇒ Differences and similarities ⇒ Plane polarisation of e.m. waves ⇒ Malus' law	AB1	State characteristic values for the wavelengths of the different regions of the electromagnetic spectrum.
	AB2	State that all electromagnetic waves travel at the same speed in a vacuum.
	AB3	Describe differences and similarities between different regions of the electromagnetic spectrum.
	AB4	Describe the plane polarisation of electromagnetic waves.
	AB5	Use Malus' law $I = I_0 \cos^2 \theta$ to calculate the intensity of a plane polarised wave after transmission through a polarising filter or filters.

TOPIC AC – WAVES: REFLECTION, DIFFRACTION AND REFRACTION		
CONTENT	SLOs	
⇒ Electromagnetic waves ⇒ Reflection of waves ⇒ Diffraction of waves ⇒ Refraction of waves ⇒ Dispersion of colours	AC1	Describe reflection of plane waves from plane surfaces.
	AC2	Describe the difference between polarised and unpolarised light.
	AC3	Explain diffraction in terms of secondary emitters of waves along a wavefront.
	AC4	Describe the effect of gap width on a diffraction pattern.
	AC5	Explain the bending of light in a lens in terms of refraction.
	AC6	Explain dispersion of colours in a prism in terms of different refractive indices for light of different wavelengths.
	AC7	Describe experiments that demonstrate reflection, diffraction and refraction of visible light, microwaves and water waves.

TOPIC AD – WAVES: INTERFERENCE		
CONTENT	SLOs	
⇒ Interference ⇒ Constructive and destructive interference ⇒ Young's double slit experiment ⇒ Diffraction grating	AD1	State and use the principle of superposition of waves and apply graphical methods to illustrate this principle.
	AD3	Explain the terms <i>interference</i> , <i>coherence</i> , <i>path difference</i> and <i>phase difference</i> .
	AD4	Describe constructive interference and destructive interference in terms of path difference and phase difference.
	AD5	Describe experiments that demonstrate two-source interference using sound, light and microwaves.
	AD6	Describe the Young double-slit experiment and explain how it is a classical confirmation of the wave-nature of light.
	AD7	Select and use the equation $w = \lambda D/s$ for the fringe spacing.
	AD8	Describe the use of a diffraction grating to determine the wavelength of light.
	AD9	Select and use the equation: $\Rightarrow d \sin \theta = n\lambda$

TOPIC AE – WAVES: STATIONARY WAVES		
CONTENT	SLOs	
⇒ Stationary waves ⇒ Nodes and antinodes ⇒ Standing wave patterns ⇒ Modes of vibration ⇒ Measurement of speed of sound using a resonance tube	AE1	Explain the formation of stationary (standing) waves using graphical methods.
	AE2	Define the terms <i>node</i> and <i>antinode</i> .
	AE3	Describe experiments to demonstrate stationary waves using microwaves, stretched strings and air columns.
	AE4	Describe the standing wave patterns for a stretched string between two fixed supports and for air columns in closed and open pipes.
	AE5	Define and use the terms for modes of vibration, <i>fundamental</i> and <i>harmonic</i> .
	AE6	Describe experiments to determine the speed of sound in air from measurements on stationary waves in a pipe closed at one end.

TOPIC AF – ATOMS: THE NUCLEAR ATOM		
CONTENT	SLOs	
⇒ The nuclear atom ⇒ Rutherford alpha-particle scattering ⇒ Basic atomic structure ⇒ Strong nuclear force ⇒ Density of nucleus ⇒ Proton and nucleon numbers ⇒ Isotopes ⇒ Simple decay equations	AF1	Describe how the Rutherford alpha-particle scattering experiment provides evidence for the existence, charge and small size of the nucleus.
	AF2	Describe the basic atomic structure of the atom and the relative sizes of the atom and the nucleus.
	AF3	Describe how the strong nuclear force between nucleons is attractive and very short-ranged.
	AF4	Know how to form an estimate of the density of the nucleus.
	AF5	Define proton number Z and nucleon number A.
	AF6	State and use the notation A_ZX for the representation of nuclides.
	AF7	Define and use the term <i>isotopes</i> .
	AF8	Use nuclear decay equations to represent simple nuclear reactions.
	AF9	State the quantities conserved in a nuclear decay.

TOPIC AG – ATOMS: RADIOACTIVITY		
CONTENT	SLOs	
⇒ Radioactivity ⇒ α and β-particles and γ-rays ⇒ Activity and decay constant ⇒ Half-life ⇒ Uses of radioactive isotopes ⇒ Decay of unstable nuclei	AG1	Describe the spontaneous and random nature of radioactive decay of unstable nuclei.
	AG2	Describe the nature, penetration and range of α -particles, β -particles and γ -rays. (Both β^- (electrons) and β^+ (positrons) are included).
	AG3	Describe practical methods to distinguish between α -particles, β -particles and γ -rays.
	AG4	Define and use the quantities <i>activity</i> and <i>decay constant</i> .
	AG5	Select and apply the equation for activity: ⇒ $A = \lambda N$
	AG6	Select and apply the equations: ⇒ $A = A_0 e^{-\lambda t}$ ⇒ $N = N_0 e^{-\lambda t}$
	AG7	Define the term <i>half-life</i> .
	AG8	Select and use the equation: ⇒ $\lambda t_{1/2} = \ln 2$
	AG9	Use graphs of $\ln(A)$ or $\ln(N)$ against t to determine λ .
	AG10	Describe the use of radioactive isotopes in smoke alarms.
	AG11	Describe the technique of radioactive (e.g. carbon) dating.

TOPIC AH – ATOMS: NUCLEAR FISSION AND FUSION		
CONTENT	SLOs	
⇒ Nuclear fission and fusion ⇒ Einstein's equation ⇒ Binding energy ⇒ Induced nuclear fission ⇒ Chain reactions ⇒ Fission reactors ⇒ Nuclear fusion	AH1	State and use Einstein's equation: ⇒ $\Delta E = \Delta mc^2$
	AH2	Define <i>binding energy</i> and <i>binding energy per nucleon</i> .
	AH3	Use and interpret the binding energy per nucleon against nucleon number graph.
	AH4	Determine the binding energy of nuclei using $\Delta E = \Delta mc^2$ and masses of nuclei.
	AH5	Describe the process of induced nuclear fission.
	AH6	Describe and explain the process of a nuclear chain reaction.
	AH7	Describe the basic construction of a fission reactor and explain the role of the fuel rods, control rods and the moderator.
	AH8	Describe the use of nuclear fission as an energy source.
	AH9	Describe the process of nuclear fusion.
	AH10	Calculate the energy released in nuclear reactions for given masses of the initial material, making use of the concept of the <i>mole</i> .

TOPIC AI – ATOMS: PHOTONS		
CONTENT	SLOs	
⇒ Photons ⇒ Photon energy ⇒ The electron volt ⇒ Line spectra	AI1	Describe the particulate nature (photon model) of electromagnetic radiation.
	AI2	Select and use the equations for the energy of a photon: ⇒ $E = hf$ ⇒ $E = hc/\lambda$
	AI3	Use the <i>electron volt</i> (eV) to express small energies.
	AI4	Explain line spectra in terms of photon emissions due to electronic transitions between atomic energy levels.

TOPIC AJ – ATOMS: THE PHOTOELECTRIC EFFECT		
CONTENT	SLOs	
⇒ The photoelectric effect ⇒ Work function and threshold frequency ⇒ Photoelectric equation	AJ1	Describe and explain the phenomenon of the photoelectric effect.
	AJ2	Define and use the terms <i>work function</i> and <i>threshold frequency</i> .
	AJ3	State that energy is conserved when a photon interacts with an electron.
	AJ4	Select, explain and use Einstein's photoelectric equation: ⇒ $hf = \phi + E_{k\max}$
	AJ5	Explain why the maximum kinetic energy of photoelectrons is independent of intensity and why the photoelectric current in a photocell circuit is proportional to intensity of the incident radiation.

TOPIC AK – ATOMS: WAVE-PARTICLE DUALITY		
CONTENT	SLOs	
⇒ Wave-particle duality ⇒ Wave-like nature of electrons ⇒ De Broglie equation	AK1	Explain how the photoelectric effect provides evidence for a particulate nature of electromagnetic radiation while phenomena such as interference and diffraction provide evidence for a wave nature.
	AK2	Explain electron diffraction as evidence for the wave nature of particles like electrons.
	AK3	Explain that electrons travelling through polycrystalline graphite will be diffracted by the atoms and the spacing between the atoms.
	AK4	Select and apply the de Broglie equation: ⇒ $\lambda = h/(mv)$.
	AK5	Explain that the diffraction of electrons by matter can be used to determine the arrangement of atoms and the size of nuclei.

TOPIC AL – ATOMS: ENERGY LEVELS IN ATOMS		
CONTENT	SLOs	
⇒ Energy levels in atoms ⇒ Emission and absorption spectra	AL1	Explain how spectral lines are evidence for the existence of discrete energy levels in isolated atoms, i.e. in a gas discharge lamp.
	AL2	Describe the origin of emission and absorption line spectra.
	AL3	Use the relationships: ⇒ $hf = E_1 - E_2$ ⇒ $hc/\lambda = E_1 - E_2$

APPENDIX A - EXEMPLAR TEACHING PLAN

Week	Hours	Topic	
1	1	A - Physical quantities and units	Lecture: ⇒ Explain that physical quantities have a magnitude and a unit. ⇒ Use and express SI units and symbols correctly. ⇒ Physical quantities should be written in italic type, units in Roman (upright) type. Both quantities and units may be written in full or using the approved abbreviation.
	1	A - Prefixes & Subfixes	Seminar/Tutorial: ⇒ Use prefixes and subfixes to indicate decimal multiples and submultiples of units. Leave a space between different units, but not between prefixes or suffixes and a unit. (mms^{-1} is incorrect. Is it meant to be: mm s^{-1} or m ms^{-1} ?)
	1	B - Scalars and vectors	Lecture: ⇒ Explain that scalar quantities have magnitude and vector quantities have magnitude and direction. ⇒ Draw and use vector triangles to determine the resultant of two co-planar vectors.
	1	B - Scalars and vectors	Seminar/Tutorial: ⇒ Calculate the resultant of two co-planar vectors and resolve vectors into two perpendicular components.
2	1	C - Kinematics	Lecture/Tutorial: ⇒ Plot graphs to represent displacement, velocity and acceleration. Determine these quantities from the gradients and areas of graphs. ⇒ Students should learn to correctly label axes and plot graphs, both by hand and using computer software.
	1	C - Kinematics	Lecture/Seminar: ⇒ Select and use the equations of motion for constant acceleration. ⇒ Determine the equation in one dimension from a velocity-time graph. This will include the motion of projectiles. The equations of motion should be applied separately to movement in the horizontal and vertical directions. ⇒ Describe a "free fall" experiment to determine g.
	1	D - Dynamics	Lecture/Tutorial: ⇒ Use the equation $F = ma$ and define the newton.

			⇒ Use this equation and the equations of motion to solve problems. ⇒ Explain the concept of weight, $W = mg$. Weight of an object is the gravitational force acting on the object.
	1	D - Friction and Drag	Seminar/Tutorial: ⇒ Explain how friction and drag may affect the motion of objects in both one and two dimensions. ⇒ Explain terminal velocity.
3	1	E - Newton's laws of motion	Lecture/Tutorial: ⇒ State and use all three laws.
	1	E - Momentum	Seminar/Tutorial: ⇒ Define linear momentum and explain how net force equals rate of change of momentum.
	1	F - Statics	Lecture/Tutorial: ⇒ Explain centre of gravity and describe simple experiments to determine it. ⇒ Define the moment of a force and apply the principle of moments to solve problems. Single support, two support and cantilever systems in equilibrium should all be considered.
	1	F - Couples and torques	Lecture/Tutorial: ⇒ Define and apply the terms couple and torque. ⇒ Draw and use a triangle of forces to solve problems of the equilibrium of coplanar forces.
4	1	G - Work and conservation of energy	Lecture/Tutorial: ⇒ Define work and the joule. ⇒ Calculate work done by a force. Solve problems involving both: ○ $W = Fx$ ○ $W = Fx \cos\theta$
	1	G - Conservation of energy	Lecture/Tutorial: ⇒ State and apply the principle of conservation of energy. ⇒ Solve problems involving work done and energy transfer. Include the pressure equation $p = F/A$
	1	H - Kinetic and potential energies	Lecture/Tutorial: ⇒ State and use the equation for kinetic energy. ⇒ Define potential energy. ⇒ Use the law of conservation of mechanical energy in problems for motion under gravity. Gravitational potential energy of a mass near the Earth's surface = mgh

	1	H - Power and efficiency	Lecture/Tutorial: ⇒ Define power and the watt and solve problems involving power. ⇒ Use the relationship for efficiency and explain how it is always less than 100% due to heat losses.
5	1	I - Conservation of momentum	Lecture/Tutorial: ⇒ State the principle of conservation of momentum and apply it to problems in one dimension.
	1	I - Collisions	Lecture/Tutorial: ⇒ Define perfectly elastic and inelastic collisions. ⇒ Explain how momentum is always conserved, but kinetic energy often changes.
	1	J - Circular motion	Seminar/Tutorial: ⇒ Define the radian and convert angles between degrees and radians and vice versa. Explain why a body moving in a circular path at constant speed is still accelerating.
	1	J - Circular motion	Lecture/Tutorial: ⇒ Explain centripetal acceleration and force. ⇒ Select and use the equations of circular motion. ⇒ Use the equations for calculating velocity and acceleration.
6	1	K - Simple Harmonic Motion	Lecture/Tutorial: ⇒ Define and use SHM and associated terms. ⇒ Determine the period of the motion. ⇒ Use the SHM equations. Use equations to calculate acceleration, displacement and velocity of a particle during SHM.
	1	K - SHM	Tutorial: ⇒ Continue with the above calculations.
	1	K - SHM	Lecture/Seminar: ⇒ Explain how period is independent of amplitude during SHM. ⇒ Describe with graphical illustrations how displacement, velocity and acceleration change during SHM. ⇒ Describe the interchange of potential and kinetic energy. Use graphs to show this. ⇒ Discuss the effect of damping on SHM.
	1	K - Resonance	Seminar: ⇒ Describe and discuss resonance. ⇒ Use graphs to show how the amplitude of a forced oscillation changes near its natural frequency.

7	1	L - Solid, liquid and gas	Lecture/Tutorial: ⇒ Describe these states in terms of the spacing, ordering and motion of atoms and molecules. Explain density. Use graphs of temperature against energy to show a change of state. Describe the simple kinetic model of gases. ⇒ Use equations for density, $\rho = M/V$ and hydrostatic pressure, $\Delta p = \rho g \Delta h$.
	1	L - Brownian motion	Seminar/Tutorial: ⇒ Describe Brownian motion. ⇒ Define pressure and use the kinetic model to explain the pressure exerted by a gas. ⇒ Describe the smoke cell and/or the motion of pollen grains as evidence for the movement of particles.
	1	M - Elasticity	Lecture/Tutorial: ⇒ Describe tensile and compressive deformation of solids. Describe the behaviour of springs and wires when stretched. Define terms associated with elastic behaviour and perform relevant calculations. Cover force, extension, elastic limit and Hooke's law. ⇒ Apply $F = -kx$. Include stress, strain, Young's modulus, elastic limit and tensile strength.
	1	M - Elasticity	Seminar/Tutorial: ⇒ Continue with the above. ⇒ Select and use the equations $E_p = \frac{1}{2}Fx = \frac{1}{2}kx^2$ for a material deformed within its limit of proportionality. ⇒ Describe an experiment to determine Young's modulus of a wire. ⇒ Describe the features of the graph of force against extension for ductile materials. Include elastic and plastic deformation.
8	1	N - Thermal properties of materials.	Lecture/Tutorial: ⇒ Define specific heat capacity and do appropriate calculations. ⇒ Describe an experiment to determine the specific heat capacity of a solid or a liquid. ⇒ Use $E = mc\Delta\theta$
	1	N - Thermal properties of materials	Seminar/Tutorial: ⇒ Describe latent heats of fusion and vaporisation, discuss the energy changes involved and do appropriate calculations. ⇒ Use $E = L\Delta m$
	1	O - Thermal physics	Lecture/Seminar: ⇒ Define internal energy of a system and explain how it changes as the temperature of a body is increased. ⇒ Explain how a change in state affects internal energy, but not the temperature of the body.

			⇒ Explain how thermal energy is transferred from regions of higher to lower temperatures.
	1	O - Thermal physics	Lecture/Seminar: ⇒ Explain that regions of equal temperature are in thermal equilibrium. ⇒ Explain the thermodynamic (Kelvin) temperature scale as an absolute temperature scale. State that absolute zero is the temperature at which a substance has a minimum internal energy. ⇒ Convert temperatures between Celsius and Kelvin scales. Use $T(K) = \theta(^{\circ}C) + 273.2$
9	1	P - Ideal gases	Lecture/Tutorial: ⇒ State and use Boyle's law, $pV = \text{constant}$ ⇒ State and use the ideal gas equation. Use
	1	P - Ideal gases	Lecture/Seminar: ⇒ State and use the basic assumptions of the kinetic theory of gases. Interpret the pressure of a gas in terms of molecular collisions and its temperature in terms of molecular energies. ⇒ Basic assumptions: ○ Intermolecular forces are negligible except during a collision. ○ The volume of molecules is negligible compared with that of the container. ○ The time occupied by a collision is negligible compared with the time between them. ○ Between collisions a molecule moves with a uniform velocity.
	1	P - Ideal gases	Lecture/Tutorial: ⇒ Explain the Avogadro constant and the definition of 1 mole of a substance. Select and use the equation: ○ $pV = nRT$ and $pV = NkT$.
	1	P - Ideal gases	Tutorial: ⇒ Continue calculations based on ideal gases.
10	1	Q - Charge and current	Lecture: ⇒ Explain current as a net flow of charged particles. Explain how current in a metal is a flow of electrons and in an electrolyte the movement of ions. ⇒ Explain conventional current and electron flow. A quantitative study of drift velocity is not required. Conventional current flows from positive to negative.
	1	Q - Charge and current	Lecture/Tutorial: ⇒ Define the coulomb and use the equation $\Delta Q = I\Delta t$. ⇒ Describe the use of an ammeter to measure current. The SI definition of the ampere is not required.

	1	R - Resistance	Lecture/Tutorial: ⇒ Define resistance and the ohm. ⇒ State and use Ohm's law. ⇒ Describe the I-V characteristics of a resistor at constant temperature, a filament lamp and an LED.
	1	R - Resistivity	Lecture/Tutorial: ⇒ Describe an experiment to obtain the above I-V characteristics. ⇒ Define resistivity and perform calculations on resistivity. Use the equation $R = \rho L/A$
11	1	S - Capacitance	Lecture/Tutorial: ⇒ Define capacitance and the farad ⇒ Use $Q = CV$. ⇒ Explain how the area under a p.d. against charge graph gives the energy stored in a capacitor. Use equations to calculate the energy stored in a capacitor: ○ $W = \frac{1}{2}QV = \frac{1}{2}CV^2$.
	1	S - Capacitor discharge	Lecture/Tutorial: ⇒ Sketch graphs to show the variation with time of p.d, charge and current for a capacitor discharging through a resistor. ⇒ Define and use the time constant of a CR circuit.
	1	S - Capacitor discharge	Lecture/Tutorial: ⇒ State and use the exponential discharge equations for a capacitor.
	1	S - Capacitor discharge	Seminar/Tutorial: ⇒ Produce straight line graphs of $\ln(Q/C)$, $\ln(V/V)$ or $\ln(I/A)$ against t to determine the time constant, τ.
12	1	T - E.m.f. and p.d.	Lecture: ⇒ Define p.d. and the volt. ⇒ Describe the use of a voltmeter to measure p.d.
	1	T - E.m.f. and p.d.	Lecture/Seminar: ⇒ Define e.m.f. of a power supply. Discuss the difference between e.m.f. and p.d. ⇒ Explain all sources of e.m.f. have internal resistance. ⇒ Introduce equations for e.m.f. Include both terminal p.d. on open circuit and the energy converted into electrical energy when unit charge passes.
	1	T - E.m.f. and p.d.	Tutorial: ⇒ Perform calculations relating to the above. ⇒ $e.m.f. = I(R + r) = V + Ir$

	1	U - Power in electric circuits	Lecture/Tutorial: ⇒ Use the equations for power and the equations for electrical work: ○ $P = IV = I^2R = V^2/R$ ○ $W = IVt = QV$
13	1	V - D.C. circuits	Lecture/Tutorial: ⇒ Draw and interpret circuits using current SI standards. ⇒ Carry out calculations on simple circuits. Use correct symbols for circuit components.
	1	V - D.C. circuits	Lecture/Tutorial: ⇒ State Kirchhoff's 1st and 2nd laws and apply the laws to circuits. Describe the 1st law as a consequence of conservation of charge and the 2nd law as a consequence of conservation of energy.
	1	V - D.C. circuits	Lecture/Tutorial: ⇒ Perform calculations on resistors in series and in parallel. ⇒ Explain the potential divider circuit for producing a variable p.d. Use the potential divider equation. ⇒ Relate V_{out} to V_{in} .
	1	V - Capacitors in D.C. circuits	Lecture/Tutorial: ⇒ Perform calculations on capacitors in series and in parallel. ⇒ Solve circuit problems with one or more sources of e.m.f.
14	4	Revision	
15	End of Semester 1 Test		
16	1	W - Electric fields	Lecture/Tutorial: ⇒ State that these are created by electric charges. ⇒ Define electric field strength and use the equation $E = F/q$. Use the equation for Coulomb's law.
	1	W - Electric fields	Lecture/Tutorial: ⇒ State and use the equation for electric field strength of a point charge. ⇒ Define and use the equation for electric potential at a point.
	1	W - Electric fields	Seminar/Tutorial: ⇒ Consider the uniform electric field between oppositely charged parallel plates. ⇒ Sketch electric field lines and equi-potentials for point charges, spherically symmetric charge distributions, the region between parallel plates and an electric dipole.

	1	W - Electric fields	Lecture/Tutorial: ⇒ Explain how a particle with charge q is accelerated by a p.d. ΔV and gains kinetic energy such that $q\Delta V = \frac{1}{2}mv^2$. ⇒ In this section, it is possible to combine the values of the velocity and acceleration with the equations of motion to analyse the deflection of a particle moving through an electric field.
17	1	X - Magnetic fields	Lecture/Tutorial: ⇒ Define magnetic field or magnetic flux density in terms of the force on a current carrying conductor. ⇒ Apply Fleming's left-hand rule.
	1	X - Magnetic fields	Lecture/Tutorial: ⇒ Select and use the equations for the force on a current carrying conductor.
	1	X - Magnetic fields	Lecture/Tutorial: ⇒ Select and use the equations for the force on a charged particle moving perpendicular to a magnetic field. Relate this to circular motion. (This allows some revision of circular motion). The motion of a charged particle moving through mutually perpendicular electric and magnetic fields can be analysed.
	1	X - Magnetic fields	Seminar/Tutorial: ⇒ Sketch magnetic field lines for a long straight current carrying wire, a long solenoid, a bar magnet and the Earth.
18	1	Y - Electromagnetism	Lecture/Tutorial: ⇒ Investigate the torque on a rectangular coil placed with its plane parallel to a magnetic field. Consider the operation of an iron cored d.c. motor with commutator, in a radial magnetic field. Consider situations when the perpendicular to the plane of the coil is making any angle θ with the field.
	1	Y - Electromagnetism	Seminar/Tutorial: ⇒ Define magnetic flux and the weber. ⇒ Make use of the equations for magnetic flux linkage and apply Faraday's law of electromagnetic induction.
	1	Y - Electromagnetism	Lecture/Tutorial: ⇒ State Lenz's law. Use this and Fleming's right-hand rule to determine the direction of induced currents and e.m.f.s.
	1	Y - Electromagnetism	Seminar/Tutorial: ⇒ Describe the a.c. generator and its operation.

19	1	Z - Gravity	Lecture/Tutorial: ⇒ Describe the existence of gravitational fields around any mass. ⇒ Define gravitational field strength and use field lines to represent the field. ⇒ Define and use gravitational potential at a point and introduce Newton's law of gravitation.
	1	Z - Gravity	Lecture/Tutorial: ⇒ Apply the equation for the gravitational field strength of a point mass. Determine the mass of the Earth or another body from its gravitational field strength. Consider the value of the gravitational field strength close to the Earth's surface.
	1	Z - Gravity	Lecture/Tutorial: ⇒ Analyse the circular orbits of planets and satellites.
	1	Z - Gravity	Lecture/Tutorial: ⇒ Apply Kepler's 3rd law. Define a geostationary orbit and use the concept in calculations. Show that $T^2 = 4\pi^2 r^3 / (GM)$ is the same relationship as Kepler's 3rd law.
20	1	AA - Waves	Lecture/Tutorial: ⇒ Distinguish between progressive longitudinal and transverse waves. ⇒ Define terms associated with waves. Include displacement, amplitude, wavelength, phase difference, frequency and speed. ⇒ Use the wave equation $v = f\lambda$.
	1	AB - Electromagnetic waves	Lecture/Seminar: ⇒ State characteristics for different regions of the e.m. spectrum. ⇒ Describe polarisation and how it only applies to transverse waves.
	1	AB - Polarisation	Lecture/Tutorial: ⇒ Consider reflection of waves from surfaces and the difference between polarised and unpolarised light. Only consider plane waves reflecting from plane surfaces.
	1	AB - Polarisation	Lecture/tutorial: ⇒ State and use Malus' law, $I = I_0 \cos^2 \theta$.
21	1	AC - Diffraction	Lecture/Seminar: ⇒ Explain diffraction in terms of secondary emitters of waves along a wavefront. ⇒ Discuss the effect of gap-width on the pattern.
	1	AC - Refraction	Lecture/Seminar: ⇒ Explain how refraction leads to the bending of light by a lens. Discuss the dispersion of coloured light by a

			triangular prism. A quantitative treatment of refraction is not required.
	1	AC - Reflection, diffraction and refraction	Seminar/Tutorial: ⇒ Describe experiments to demonstrate reflection, diffraction and refraction. Consider visible light, microwaves and water waves.
	1	Introduce Coursework	⇒ Introduce summative coursework. The details for this can be found in the IFY Physics Laboratory Practical Guide available from the NCUK Teachers' Hub. Note that you are free to schedule this session as suitable to your teaching calendar and the practical to be undertaken.
22	1	AD - Interference	Lecture/Tutorial: ⇒ Explain the principle of superposition. Apply graphical methods to illustrate this. This might be done using computer software to show how various waveforms can be constructed from separate sine waves.
	1	AD - Interference	Lecture/Tutorial: ⇒ Explain terms associated with wave interference. Explain constructive and destructive interference. Relate this to path difference and phase difference. ⇒ Describe experiments to demonstrate interference. Include two-source interference using sound, light and microwaves.
	1	AD - Interference	Lecture/Tutorial: ⇒ Describe Young's double slit experiment, including calculations using $w = \lambda D/s$.
	1	AD - Interference /diffraction	Lecture/Tutorial: ⇒ Describe the use of a diffraction grating and include calculations using $d \sin \theta = n\lambda$.
23	1	AE - Stationary Waves	Lecture/Tutorial: ⇒ Explain the formation of stationary waves using graphical methods. Define node and antinode. Computer software could be used.
	1	AE - Stationary Waves	Lecture/Seminar: ⇒ Describe experiments to demonstrate stationary waves. Include microwaves, stretched strings and air columns.
	1	AE - Stationary Waves	Seminar: ⇒ Describe standing wave patterns for stretched strings and air columns in both open pipes and those closed at one end. Include the separation of adjacent nodes and antinodes. ⇒ Consider the different modes of vibration. Consider fundamental and harmonic modes of vibration.

	1	AE - Stationary Waves	Lecture/Seminar: ⇒ Describe the resonance tube experiment to determine the speed of sound in air. Include the end correction of the tube in calculations.
24	1	AF - Atoms	Lecture/Seminar: ⇒ Describe the Rutherford alpha-particle scattering experiment and the resulting evidence for the basic atomic structure and properties.
	1	AF - Atoms	Lecture/Tutorial: ⇒ Describe the nature of the strong nuclear force. ⇒ Form an estimate of the density of the nucleus.
	1	AF - Atoms	Lecture/Tutorial: ⇒ Define proton number and nucleon number and use the correct notations to represent a particular nucleus. ⇒ Define the term isotope and use decay equations to represent simple nuclear reactions. State the quantities conserved in nuclear decay. Introduce the discovery of the neutron.
	1	AG - Radioactivity	Lecture: ⇒ Describe the nature of radioactive decay and the nature, penetration and range of the different radioactive particles. Include both β^+ and β^- decay and the emission of neutrinos and antineutrinos in this decay process.
25	1	AG - Radioactivity	Lecture/Seminar: ⇒ Describe methods of distinguishing between α , β and γ radiation.
	1	AG - Radioactivity	Lecture/Tutorial: ⇒ Define activity and decay constant and use the equations involving these terms.
	1	AG - Radioactivity	Lecture/Tutorial: ⇒ Define half-life and relate it to the decay constant. ⇒ Determine the half-life by calculation and graphically. Use both linear and semi logarithmic plot methods.
	1	AG - Radioactivity	Seminar: ⇒ Describe the use of radioactive isotopes in smoke alarms and radioactive dating. Detailed calculations will not be required.
26	1	AH - Nuclear fission and fusion	Lecture/Tutorial: ⇒ State and use Einstein's equation. ⇒ Define binding energy and its value per nucleon. Make use of appropriate binding energy graphs.

	1	AH - Nuclear fission and fusion	Lecture/Tutorial: ⇒ Determine binding energy of nuclei using Einstein's equation and nuclei masses. ⇒ Describe induced nuclear fission. A simple description only is required.
	1	AH - Nuclear fission and fusion	Lecture/Seminar: ⇒ Describe the idea of a chain reaction. ⇒ Describe the basic construction of a fission reactor and the use as an energy source. Include the role of the fuel rods, control rods and moderator.
	1	AH - Nuclear fission and fusion	Lecture/Tutorial: ⇒ Describe nuclear fusion. ⇒ Calculate the energy released in nuclear reactions, making use of the concept of the mole. A simple description only is required.
27	1	AI & AJ - Photons and the photoelectric effect	Lecture/Tutorial: ⇒ Describe the photon model of e.m. radiation. Make use of the equations for photon energy. ⇒ Introduce the electron volt. ⇒ Explain atomic line spectra as photon emissions due to transitions of electrons between energy levels. More details will be introduced in the final class.
	1	AI & AJ - Photons and the photoelectric effect	Lecture/Tutorial: ⇒ Describe and discuss the photoelectric effect and the photoelectric equation. Explain why there is a maximum photon kinetic energy, independent of intensity.
	1	AK - Wave-particle duality	Seminar/Lecture ⇒ Discuss how the photoelectric effect provides evidence of the particulate nature of e.m. radiation, but other phenomena provide evidence of its wave nature. ⇒ Explain electron diffraction as evidence of the wave nature of electrons and describe their diffraction in polycrystalline materials.
	1	AK - De Broglie equation	Lecture/Tutorial: ⇒ Use the de Broglie equation to determine the wavelength of electrons or other moving particles. ⇒ Explain the use of electron diffraction to determine atomic size and arrangement.
28	2	AL – Energy Levels in atoms	Lecture/Tutorial: ⇒ Explain how spectral lines are evidence of discrete energy levels in isolated atoms and the origin of emission and absorption line spectra. ⇒ Use the relationships $hf = hc/\lambda = E_1 - E_2$. ⇒ Calculate energies in joules and electron volts.

	2	Revision	
29	4	Revision	
30	Exam Week		

APPENDIX B - DATA, FORMULAE & RELATIONSHIPS

Refer to separate document “Data, Formulae and Relationships”.

Syllabus Update History: NCUK International Foundation Year Physics

Release Date	Version No.	Summary of Main Changes	Author
June 2024	4.3	Updated for 2024/25	Product Team
June 2023	4.2	Updated for 2023/24	Product Team
May 2022	4.1	Updated for 2022/23	Product Team
June 2022	4.0	Full Review	John Bunning & Product Team

This syllabus is valid for the 2024-25 academic year only.

NCUK – The University Consortium

WeWork, No1 Spinningfields, Quay Street, Manchester, M3 3JE, United Kingdom

Tel: +44 (0)161 549 9220

www.ncuk.ac.uk

©NCUK Ltd.

NCUK
THE UNIVERSITY CONSORTIUM